Robot Direction Control

1. Experiment Purpose

This section teaches you how to control the movement direction of the robot.

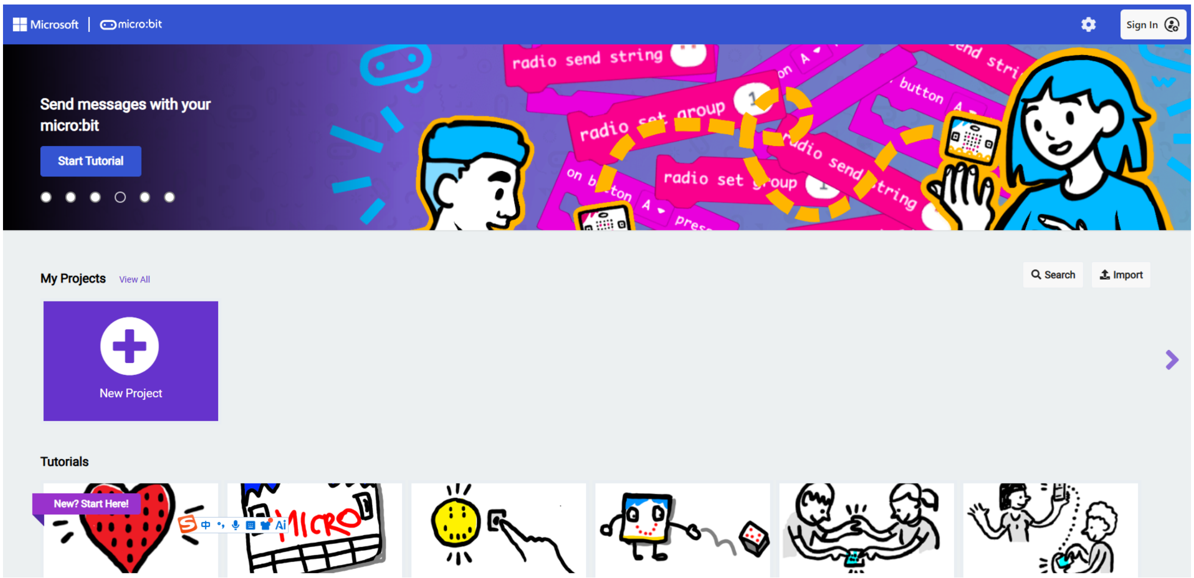
2. Experiment Steps

The blocks in this chapter belong to the `Motor Control` category under `Lastbit Motion Control`.

1. Before the experiment, use a Micro USB data cable to connect the Micro:bit main board to the computer, and the computer will display the MICROBIT drive.

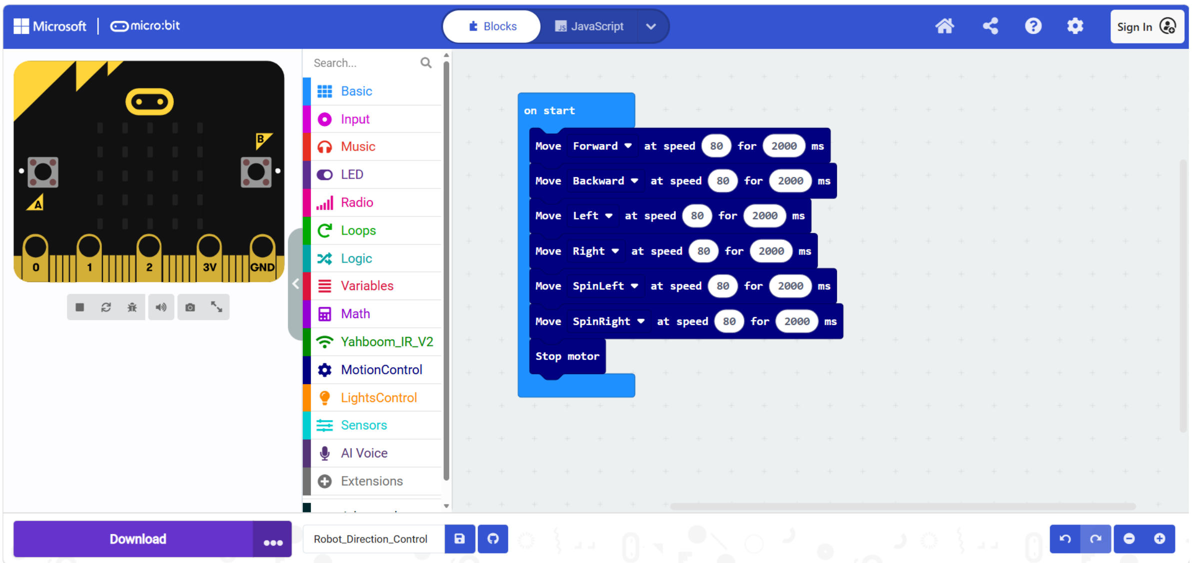
2. Open MakeCode through the link below.

[Microsoft MakeCode for micro:bit]



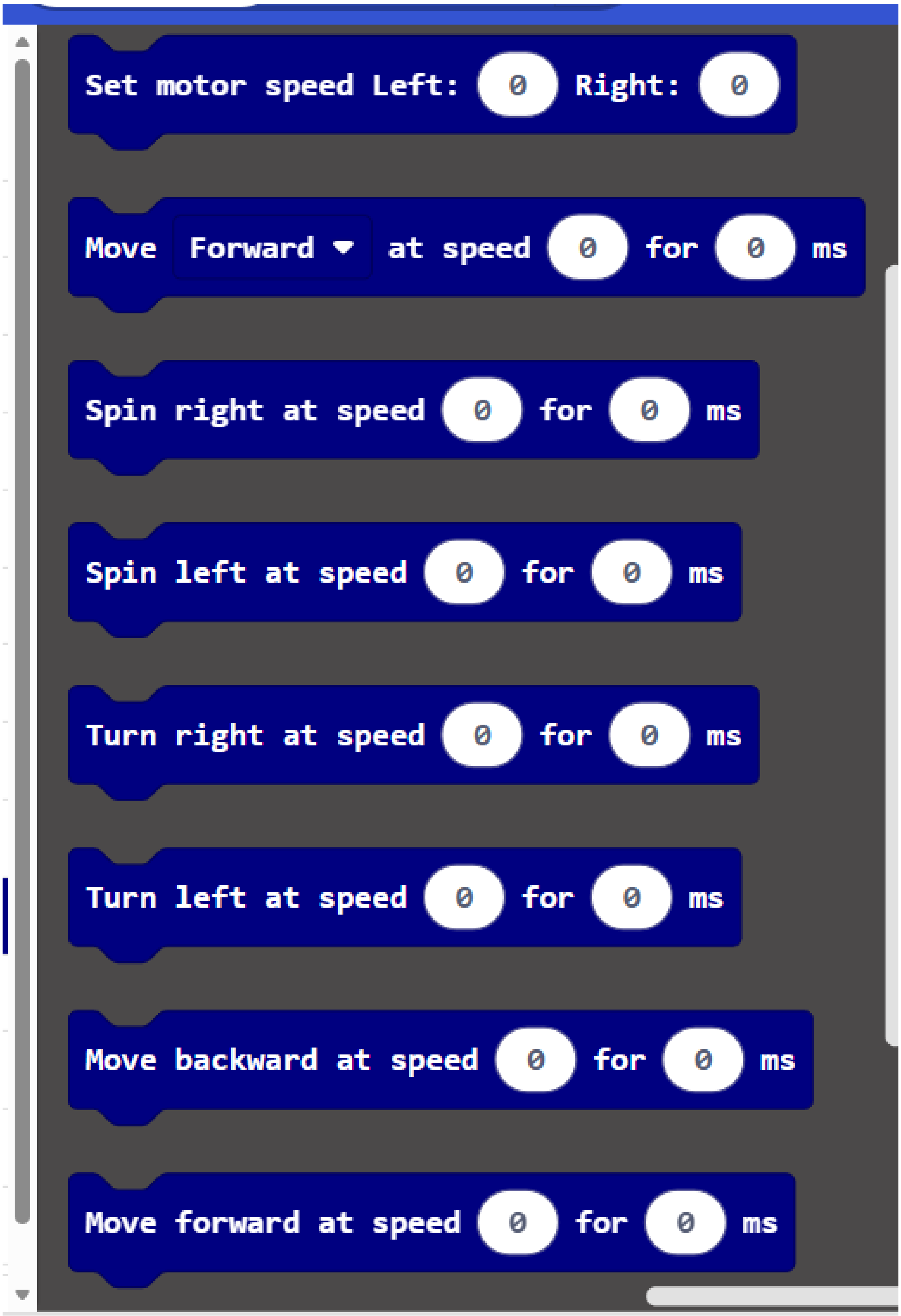
3. Drag `2.Robot_Direction_Control.hex` to the New Project page, and MakeCode will automatically import and open the project.

3. Experiment Source Code



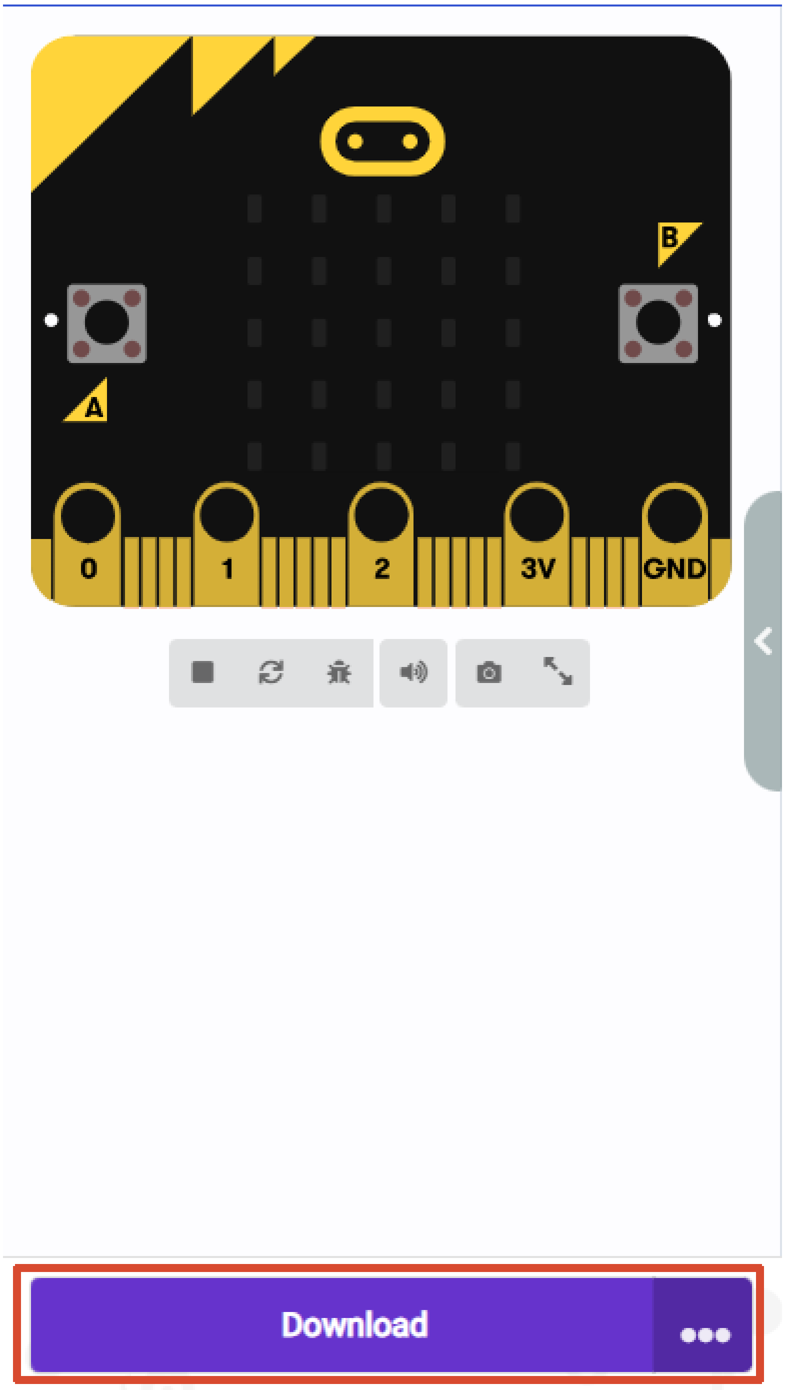
The blocks used in this tutorial are the same as those in the previous section, except that different movement directions are set here and time parameters are added for control.

In addition to the methods provided in the example, you can also use the following methods in `Motor Control` to control the car's movement direction. If you are interested, you can replace them yourself.

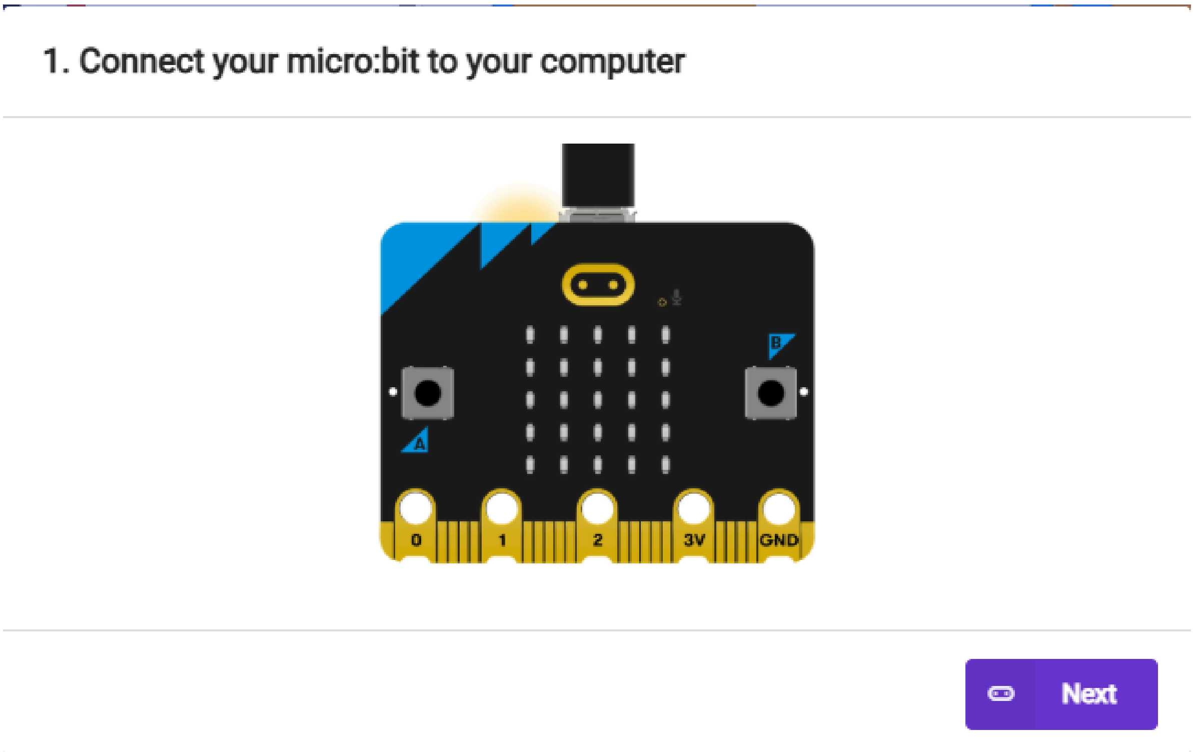


4. Program Download

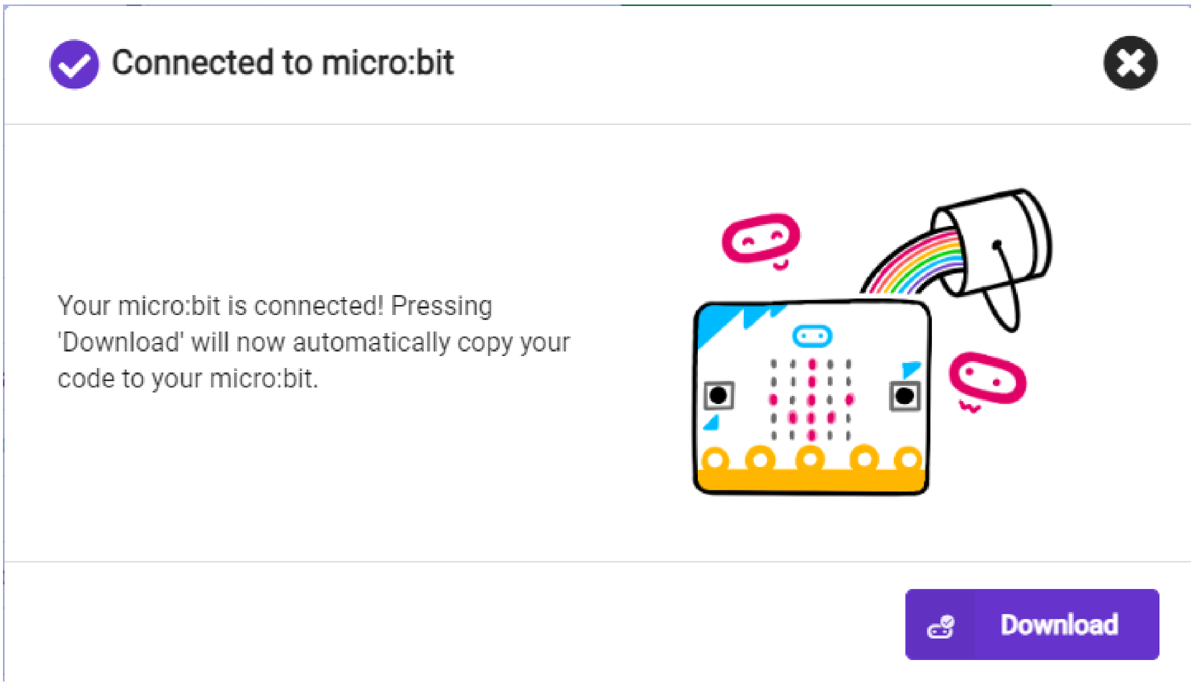
1. Click Download in the lower left corner to download the program to the Micro:bit.



2. When the Micro USB data cable is not connected to the Micro:bit, the system will first prompt you to connect the device. If it is not connected, connect the data cable to the Micro:bit first; if it is already connected, click [Next].

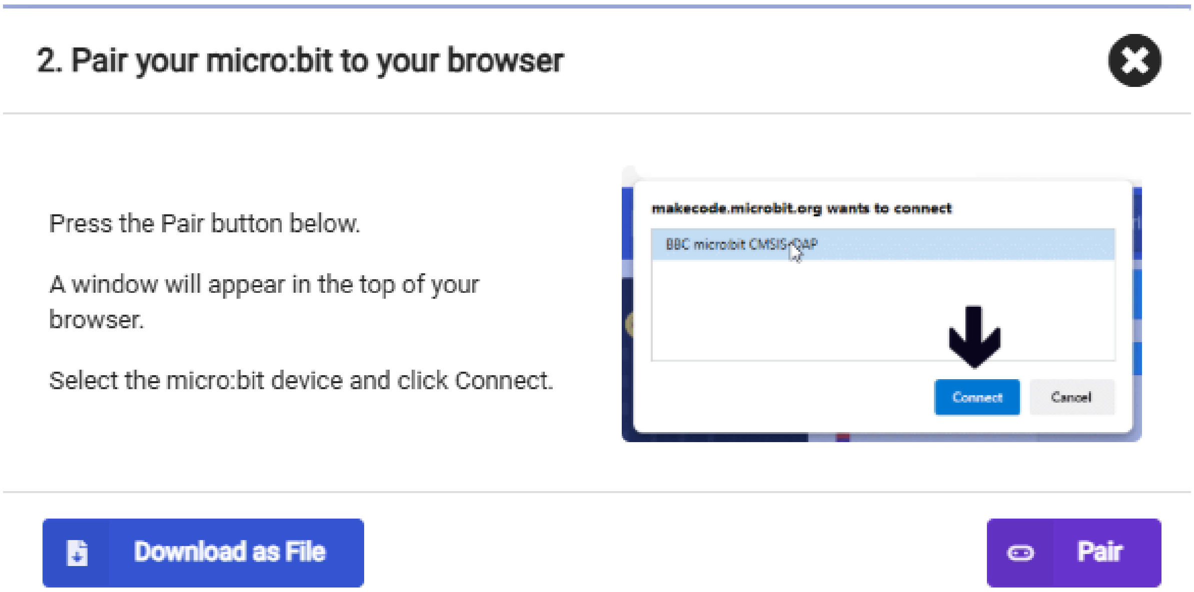


3. If the Micro:bit is detected to be connected normally, the following interface will be displayed, and you can click [Download].



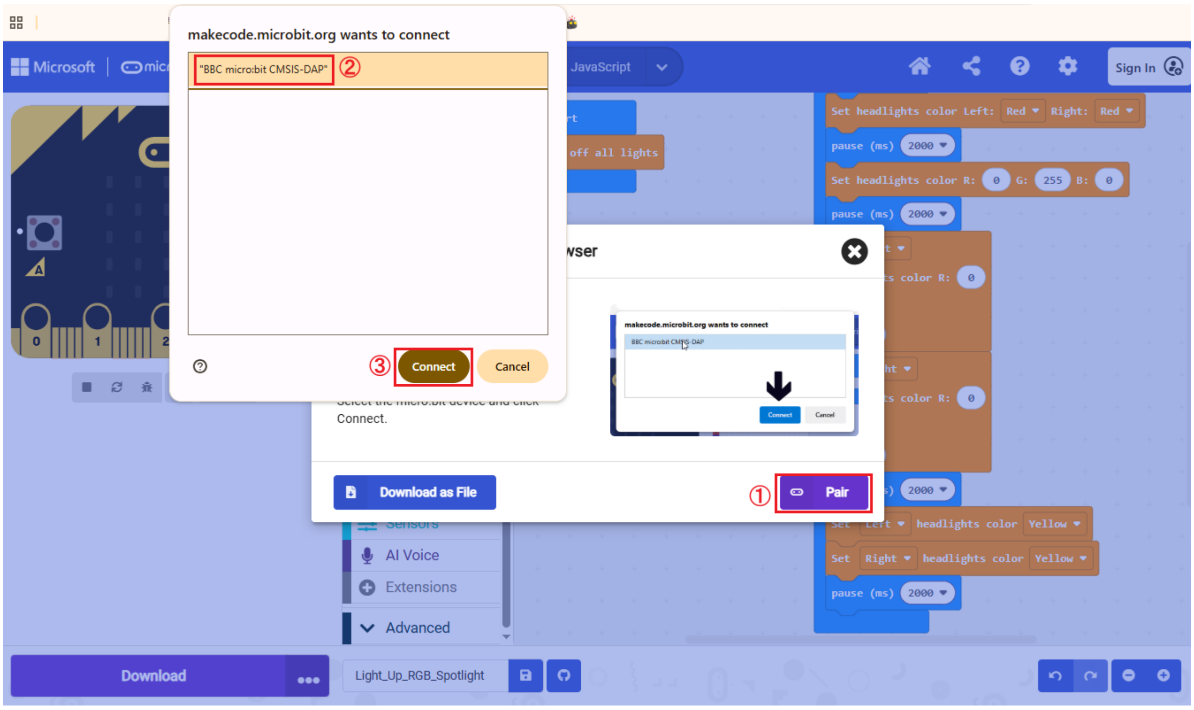
If the Micro:bit data cable is not properly connected or multiple Micro:bits are connected, the system will ask you to manually select the device. You can choose one of the following two methods:

- [Download as File]
- [Pair]

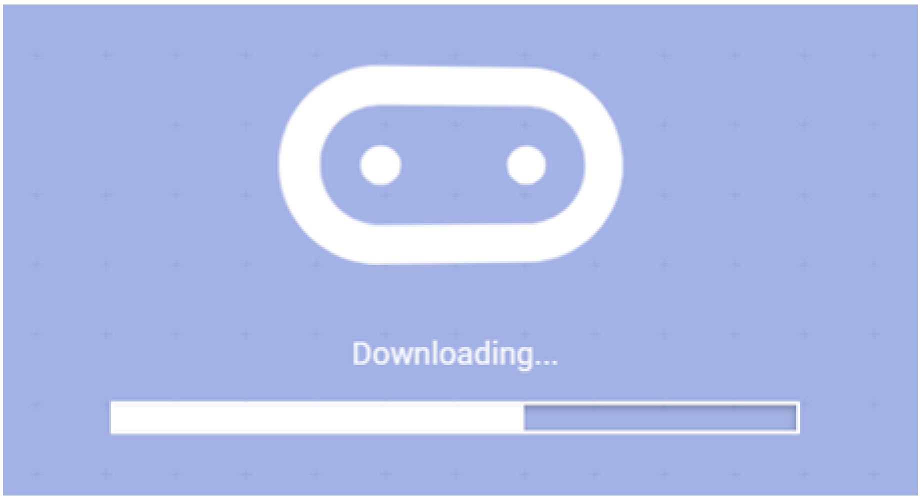


When selecting [Download as File], the system will download the currently open program as `microbit-<project name>.hex`. You can directly copy or drag this program to the corresponding MICROBIT drive to complete the download.

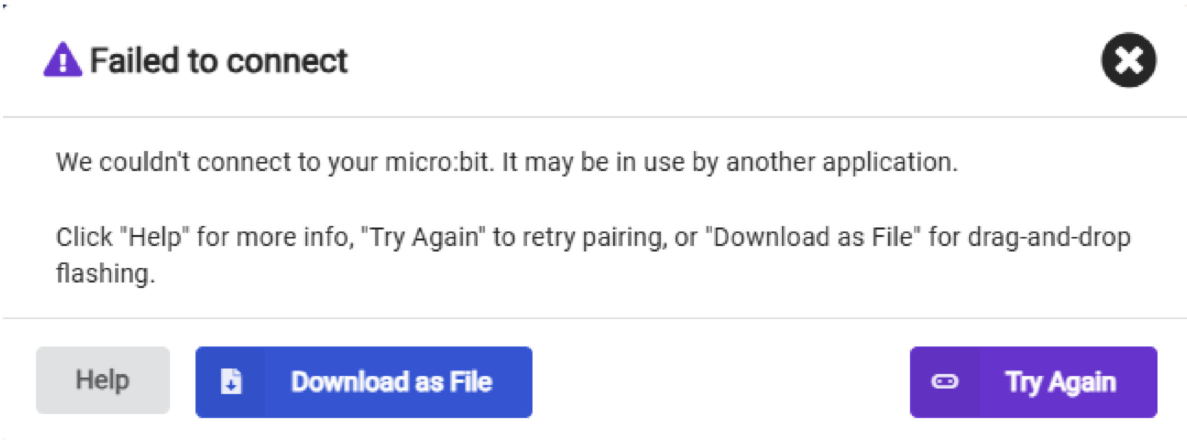
When selecting [Pair], you will be asked to manually select the device. Here, select `BBC micro:bit CMSIS:DAP`.



4. The following interface indicates that the program is being downloaded. Wait for the progress bar to finish to download the program to the Micro:bit.

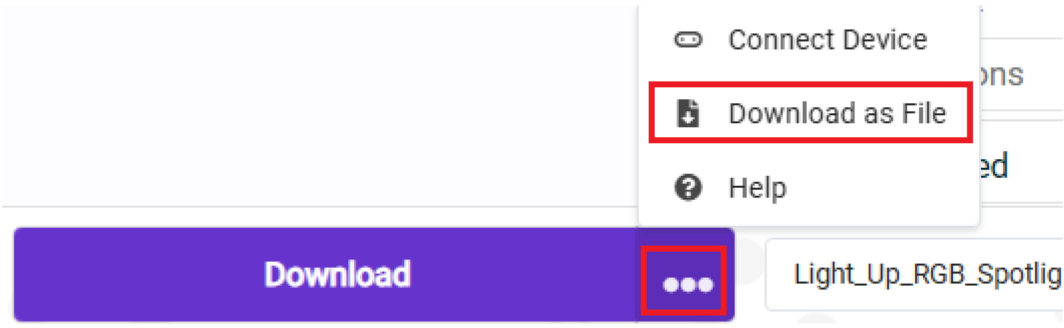


If it still fails after several attempts, please directly select [Download as File], save the current program as a file, and then copy or drag the file to the corresponding MICROBIT drive to complete the download.

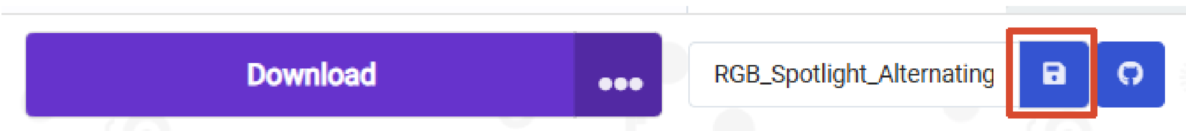


If you want to save the file directly, you can use the following two methods:

Method 1: Click [...] next to Download, and then select [Download as File].



Method 2: Click [Save] after the project name editing field.



5. Experiment Result

After flashing, turn on the power switch (toggle the switch to the ON state), and the car will sequentially perform forward -> backward -> turn left -> turn right -> rotate left -> rotate right, and finally stop.